

Real Time Prediction and Control of Transient Stability Using Transient Energy Function

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Real Time Prediction and Control of Transient Stability Using Transient Energy Function

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Abstract— Real time prediction of transient stability after a fault can potentially prevent occurrence of grid blackouts. In this work, the measurement data obtained from phasor measurement unit (PMU) located at generator buses are used to compute the transient stability margin after a fault has occurred. For evaluating various control actions to be taken to stabilize severely disturbed and unstable generators, the stability margin is estimated using the transient energy function (TEF) technique. In this work, a look-up table of various modes of disturbance (MOD) is built offline for different fault locations and post fault topology. Following an actual fault, the most probable MODs are ranked by matching of the normalised kinetic energy with the look-up table. The correct MOD is then chosen based on the lowest normalized potential energy margin and the CUEP is calculated. This technique overcomes previously reported difficulties in finding the CUEP in real-time applications. With knowledge of pre-fault operating condition obtained from SCADA and the information of the tripped line by analysing the PMU data, the first swing transient stability margin is computed. The amount of control action needed is subsequently calculated. The proposed method has been tested on the IEEE 39 Bus Test System.

Index Terms— Transient Stability, Transient Energy Function (TEF), Transient Stability Control, Generation Shedding, Dynamic Braking, Fast Valving.

I. INTRODUCTION

TRANSIENT instability is one of the less probable but most severe events that power systems encounter in daily operation. After a severe disturbance like fault, if proper control action is not taken, then one or more generators may lose synchronism with the rest. Transient instability may cause unintentional islanding, cascaded outages and even wide spread blackouts. Therefore, sufficient operating margins must be maintained at every operating point that may have significant economic implications. The ability to predict transient stability status and assess the severity of a fault in real time may allow the grid to be operated with lower transient stability margins.

Presently available methods for real time prediction of transient stability after a fault, are mostly based on ANN [1], machine learning methods like decision trees [2] and support vector machines [3] etc. In all these methods, faults at different locations of the grid are simulated offline and different dynamic features of power system measurements used to train

a classifier to enable it to distinguish between stable and unstable events. These methods were shown to provide acceptable accuracy in classification in a simulation environment. But issues related to the reliability and the practical implementation remain, which are enlisted below.

1) For a large grid, considering faults of different types, durations and fault impedances at different locations, significant time and resources are required for generating the data set for training. Considering the evolving behaviour and uncertainties associated with power system operations, the data set may need to be updated regularly, which remains a challenge.

2) Another major disadvantage of the machine learning based methods is that the training data set becomes unreliable with any change in network topology, generators' status, and major loads etc.

3) Most of these methods rely entirely on time domain simulations for building the training database. It is already well-known that time domain simulations may not always provide a qualitative assessment of fault severity. Therefore, even if the transient stability of a particular disturbance is predicted successfully, it is difficult to decide upon the most efficient control actions to be taken. For instance, in [4] a fixed amount of load shedding was proposed to stabilize the system after an unstable fault irrespective of the fault severity.

Upon successful implementation, transient energy function technique has the potential to solve the above mentioned challenges. The central premise of the TEF is that if the energy of the system at the time of fault clearing is higher than the energy at the controlling unstable equilibrium point (CUEP), the system will lose synchronism and become unstable [5]. The original problem formulation was relatively unwieldy, and subsequent improvements were proposed in the calculation of controlling unstable equilibrium point [6]-[8]. The kinetic energy of the generators at the fault clearing instant and the deceleration after fault clearing were used to identify the generators which tends to be out of step [6]-[7]. These generators are referred to as the Mode of Disturbance (MOD). The unstable equilibrium points of the generators corresponding to the MOD and the stable equilibrium points of other generators form the Controlling Unstable Equilibrium Point (CUEP). As the method is heuristic in nature, it suffers from convergence problems. A unified method was introduced to calculate the CUEP [8]. Further research was done to include the detailed generator models [9]-[10], dynamic and structure preserving load models [11], HVDC models [12] etc. Due to computational complexity, detailed models are not commonly used in

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TEF. It has been shown that classical generator models with constant load admittance model provide results with sufficient accuracy for first swing transient stability assessment [13]-[14]. In this work also, classical generator models with constant admittance loads have been used.

Despite several advantages, the real-time applications of TEF remains challenging because of several difficulties, primarily in the calculation of the CUEP [15].

1) The MOD technique suffers from serious convergence problem.

2) The Potential Energy Boundary Surface (PEBS) technique and the BCU method simulate the faulted network to find the exit point on PEBS [16, 8]. Simulating faulted network to detect the exit point needs knowledge of fault resistance, location, type and post fault topology. Estimating the fault resistance and the exact fault location in real time is difficult task especially for the BCU method which integrates the post fault gradient system starting from the exit point to find the CUEP.

In this work the aim is to predict transient stability after a fault using the PMU data. Both the MOD and the BCU technique have been used to calculate the CUEP. The MOD depends on the distribution of the kinetic energy gained by the generators during fault and the post fault topology. In the offline stage, the MOD is found out for different fault locations in the grid using the BCU (for both self clearing faults and faults cleared by breaker operation). After a fault, the normalized kinetic energy ($\overline{KE}$) of generators is compared with $\overline{KE}$ calculated offline for different fault locations to identify the most probable MODs. Finally, using the lowest ($\frac{\Delta V_{pe}}{KE_{corr}}$), the correct MOD is identified. Once the MOD is known, the CUEP and the energy margin can be computed to assess the transient stability status and estimate the amount of control action required. To identify the tripped line (post fault topology), sufficient PMUs may be deployed to make the system observable through PMU, else [17, 18] by the same authors may be followed which use PMUs at the generator buses only.

The rest of the paper is divided into six sections. The proposed method has been explained in section II. Section III presents the results obtained. The advantages of the proposed method have been explained in section IV. Section V is the concluding section.

II. TRANSIENT STABILITY ANALYSIS USING TEF

The application of TEF technique to the analysis of transient stability is classified into two approaches

A. Mode of Disturbance (MOD) Technique

After fault clearing, using the kinetic energy gained during the fault and the post-fault deceleration, the group of machines tending to lose synchronism are predicted. If machines i and j (in an m -machine system) tend to be out of step after fault clearing, MOD = $\{i, j\}$. The vector of rotor angles $[\theta_1^s \ \theta_2^s \ \theta_3^s \ \dots \ \theta_i^s \ \dots \ \theta_j^s \ \dots \ \theta_m^s]$ representing the post fault stable equilibrium point (SEP) may be computed from the post-fault steady state conditions. For this MOD, the CUEP is

$$\theta_{UMOD} = [\theta_1^s \ \theta_2^s \ \theta_3^s \ \dots (\pi - \theta_i^s) \ \dots (\pi - \theta_j^s) \ \dots \ \theta_m^s] \quad (1)$$

This angle is to be converted into the COI reference frame.

All machines are classified into two groups-

Group I: machines belong to the MOD.

Group II: rest of the machines.

$$\theta_I^s = \frac{1}{M_I} \sum_{i \in I} M_i \theta_i^s, \quad \theta_{II}^s = \frac{1}{M_{II}} \sum_{i \in II} M_i \theta_i^s \quad (2)$$

$$\theta_{I-II}^s = \theta_I^s - \theta_{II}^s$$

where, $M_I = \sum_{i \in I} M_i$, $M_{II} = \sum_{i \in II} M_i$

$$\Delta \theta_I = (\pi - 2\theta_{I-II}^s) \left(\frac{M_{II}}{M_I + M_{II}} \right) \quad (3)$$

$$\Delta \theta_{II} = (\pi - 2\theta_{I-II}^s) \left(\frac{M_I}{M_I + M_{II}} \right)$$

$$\theta_i^{u*} = \theta_i^s + \Delta \theta_I, \quad i \in I \quad (4)$$

$$\theta_i^{u*} = \theta_i^s - \Delta \theta_{II}, \quad i \in II$$

Generator angles given by θ_i^{u*} , called corner points are a good estimate of the CUEP in COI reference frame and can be made even better using Ray Technique [13]. The exact CUEP θ_i^u may be obtained by solving a optimization problem to minimize $\sum_{i=1}^m |f_i^{PF}(\theta)|$ in the post fault system.

The convergence problems of the MOD technique arises from the difficulty to determine the group of machines tending to separate after a fault with absolute certainty. Once the MOD for a fault is known, the CUEP calculation is straightforward.

B. BCU Technique

The convergence problem of the MOD method can be overcome by the BCU method that can calculate the CUEP in a unified manner-

step 1: Obtain post fault SEP by solving $f_i(\theta) = 0$.

$$M_i \frac{d^2 \theta_i}{dt^2} = P_i - \sum_{j=1, j \neq i}^n (C_{ij} \sin \theta_{ij} + D_{ij} \cos \theta_{ij}) - \frac{M_i}{M_T} P_{coi} \quad (5)$$

$$= f_i(\theta), \quad i = 1, \dots, m$$

where, $P_i = P_{mi} - E_i^2 G_{ii}$

$$P_{coi} = \sum_{i=1}^m P_i - 2 \sum_{i=1}^m \sum_{j=i+1}^m D_{ij} \cos \theta_{ij}$$

step 2: The faulted network is simulated to obtain the PEBS exit point.

$$M_i \frac{d^2 \theta_i}{dt^2} = f_i^F(\theta), \quad i = 1, \dots, m \quad (6)$$

$\sum_{i=1}^m f_i(\theta)(\theta_i - \theta_i^s)$, calculated on post fault network, becomes zero as the system reaches the PEBS boundary. Generator angles at the PEBS is represented as θ^* .

step 3: The gradient system of the post fault system ($\frac{d\theta_i}{dt} = f_i^{PF}(\theta)$) of half the dimensions of the original system (speed dynamics not considered) is integrated until $\sum_{i=1}^m |f_i(\theta)|$ becomes minimum. The exact CUEP is calculated by solving a load flow like minimization problem (step 1) with initial point at the minimum of $\sum_{i=1}^m |f_i(\theta)|$.

As the machines belonging to the MOD have higher angles at the CUEP, the MOD can be identified from the angles of the generators at the CUEP.

BCU method can't be applied in real time as integration of the faulted and the post fault network in real time is time consuming and requires knowledge of the exact fault location, fault resistance etc. However, it can be used in offline to get the CUEP that can be used for identifying the MOD.

III. PROPOSED METHOD

The proposed method relies on the fact that PMUs are placed on all the generator buses, and their measurements are available. A look-up table consisting of the MOD and normalised kinetic energy for faults at different locations is initially built from offline simulations. Here, the MOD is obtained using the BCU technique for every simulated fault. In real-time, as explained in Section II, the CUEP can be calculated if the MOD is known. Therefore, for an actual fault the normalised kinetic energy of all the machines is compared with the look-up table to rank the possible MODs for that fault. The correct MOD is then identified from the ranked MODs as the one having the minimum $\frac{\Delta V_{pe}}{KE_{corr}}$. After identifying the correct MOD, the CUEP is calculated as mentioned in Section II-A, and the transient stability margin computed. The entire exercise may be divided into two parts – the offline computation, which has two steps *a)* identification of MOD for different fault locations using BCU method, and *b)* computation of normalized KE gained by the generators for faults at different locations of the grid, and the online computation comprising of three steps – *a)* correct MOD identification after a fault, *b)* assessment of first swing transient stability, and *c)* estimation of the amount of control action required.

A. Offline Computation –building the look-up table

a) MOD identification for different fault location:

For a particular operating condition, the CUEP and the MOD depend on the fault location and the post fault topology. Therefore, for every fault simulated, the CUEP is calculated using the BCU technique. The MOD is obtained as the generators with advanced angles at the CUEP.

b) Computation of normalized KE gained by the generators:

For fault at a particular location "X" in the grid, the kinetic energy gained by all m -generators in first 0.1s of the fault is calculated as

$$KE_X = [ke_{1x} \ ke_{2x} \dots \ ke_{ix} \dots \ ke_{mx}] \quad (7)$$

where ke_{ix} denotes the kinetic energy gained by i -th generator and equals $0.5 M_i \Delta \omega_i^2$, with M_i =moment of inertia and $\Delta \omega_i$ =speed deviation of i^{th} generator. The normalized kinetic energy is the individual generator ke_{ix} divided by the sum of elements of KE_X .

$$\begin{aligned} \overline{KE}_X &= [\overline{ke}_{1x} \ \overline{ke}_{2x} \dots \ \overline{ke}_{ix} \dots \ \overline{ke}_{mx}] \\ &= \frac{[ke_{1x} \ ke_{2x} \dots \ ke_{ix} \dots \ ke_{mx}]}{\sum_{i=1}^m ke_{ix}} \end{aligned} \quad (8)$$

$$\text{Thus, } \sum_{i=1}^m \overline{ke}_{ix} = 1$$

For a fault at a location "X", if generator i and j go out of step, the MOD is $M = \{i, j\}$. For fault at location "Y", if no generator goes out of step even with sustained fault, $M = \{0\}$.

As MOD may change with large variation in operating condition, $\overline{KE}$ and MOD must be regularly calculated and stored for multiple operating conditions on the daily load curve.

B. Online Computation – using PMU data

a) Correct MOD Identification:

After a fault, the correct MOD may be selected from the stored cases as described below.

step 1: Calculation of $\overline{KE}$ - Calculate the $\overline{KE}$ from the generator speeds obtained from the PMU data.

step 2: Ranking of Probable MODs- The MOD depends on the distribution of kinetic energy gained by generators during a fault and as well as the post fault topology. Using k^{th} Nearest Neighbour (k-NN) method [28], the stored fault cases (at all operating conditions) with similar kinetic energy distribution are ranked in ascending order of the Euclidean distance "D" between the stored $\overline{KE}$ calculated for faults at different locations and measured $\overline{KE}$.

step 3: Calculation of Post-fault SEP- The correct MOD can be chosen using minimum $(\frac{\Delta V_{pe}}{KE_{corr}})$ from the most probable MODs ranked using k-NN method. To do so, the post fault SEP and the CUEP for each of the ranked MODs are calculated. Considering the tripped line is known if the system is PMU observable or following [17], the post fault SEP (θ^s) can be calculated as in step 1 of Section II-B.

step 4: Calculation of CUEP- For each ranked MOD, the CUEP is calculated as explained in Sec II-A.

step 5: Identification of the Correct MOD- Among all the MODs ranked, the first few are tested to identify the correct MOD. The lowest normalized potential energy margin $(\frac{\Delta V_{pe}}{KE_{corr}})$ identifies the correct MOD where, ΔV_{pe} is the difference between potential energy at fault clearing instant and at the CUEP. The KE_{corr} is the corrected kinetic energy of two machine equivalent of the system at fault clearing instant [13].

b) Assessment of first swing transient stability:

The expression of the energy function is well established and it is being used since last three decades.

$$\begin{aligned} V(\omega_i, \theta_i^u) = & 0.5 \sum_{i=1}^m M_i \omega_i^2 - \sum_{i=1}^m P_i (\theta_i - \theta_i^s) - \sum_{j=1}^{m-1} \sum_{i=j+1}^m [C_{ij} (\cos \theta_{ij} - \cos \theta_{ij}^s) - D_{ij} \frac{\theta_i - \theta_i^s + \theta_j - \theta_j^s}{\theta_{ij} - \theta_{ij}^s} (\sin \theta_{ij} - \sin \theta_{ij}^s)] \end{aligned} \quad (9)$$

The energy at the CUEP is called the critical energy (V_{cr}) of the system. If the energy of the system at fault clearing instant (V_{cl}) is less than the energy at the CUEP (V_{cr}), the system will remain stable, otherwise unstable.

c) Estimation of the Amount of Control Actions Required and Monitoring of Stability Status:

Prediction of generator speed and angles: If generation shedding is required to stabilize the system after a fault, the speed of the generators at the instant of shedding has to be predicted. The minimum amount of data available for this prediction corresponds to the computational delay associated with the application of the TEF technique (say $T_1 \sim 0.05$ s). This includes the iterative calculation of post fault SEP and the

CUEP. For practical implementation with online WAMS data, there are other delays associated signal latency ($T_d \sim 0.05 - 0.1$ s) and initiating control actions ($T_c \sim 0.05$ s). Therefore, data of duration at least T_1 s after the fault clearing has to be used to predict the generators' speeds and angles ($2T_d + T_c + T_1$) s after fault clearing. The prediction function must be selected to ensure optimal performance with minimum data. If larger duration data is used for prediction, the control actions would be delayed which will require more severe control action.

For the purpose of testing various prediction functions, data from a 0.05 s and 0.1 s time window were tested for prediction. The error in predicting generator angles after ($2T_d + T_c + T_1$) time is shown in Table 1. Combining the Quadratic fit and the Cubic fit yielded the best result with low average error in predicting generator angles and speeds. The prediction error is much below the permissible error in predicting angles which was taken as 10^0 in [23].

TABLE 1: AVERAGE ERROR IN PREDICTED GENERATOR ANGLES FOR DIFFERENT PREDICTION FUNCTIONS

Prediction Function	Duration of data used for prediction (s)	Instant at which data is to be predicted (s)	Average Prediction Error (Degree)
Quadratic	0-0.05 s	0.2	11
		0.25	21
		0.3	35
	0-0.1 s	0.25	11.9
		0.3	20.9
		0.35	30
Cubic	0-0.05 s	0.2	3.05
		0.25	9
		0.3	21
	0-0.1 s	0.25	6.7
		0.3	16.4
		0.35	29.4
(Quadratic fit + Cubic fit)/2	0-0.05 s	0.2	2.55
		0.25	7
		0.3	9.8
	0-0.1 s	0.25	3.2
		0.3	4.3
		0.35	6.3
Fast Generation Shedding Equipment [19]	0-0.05 s	0.2	23
		0.25	28
		0.3	35

Modification in Energy Function for Generation Shedding: If generation shedding is assumed in a plant the generator inertia, input power, and transient reactance change for the purpose of TEF technique. If P_{sh} power (pu value) is shed from i^{th} generator, then the new values are

$$H_{new} = H(1 - P_{sh})$$

$$P_{new} = P(1 - P_{sh})$$

$$X'_{dnew} = \frac{X'_d}{1 - P_{sh}}$$

Estimation of the amount of control actions required: A flowchart to estimate the control action required to stabilize the system is shown in Fig. 1. The selection of the plant for generation shedding is done on the basis of maximum sensitivity of the energy margin to generation shedding in different plants. To simplify the selection process, the generator having highest ratio of kinetic energy to output power may also be chosen. The minimum amount to be shed would be the unit

having the minimum output among all the units in the plant chosen for shedding. The energy is calculated for different amounts of shedding as shown in Fig. 1.

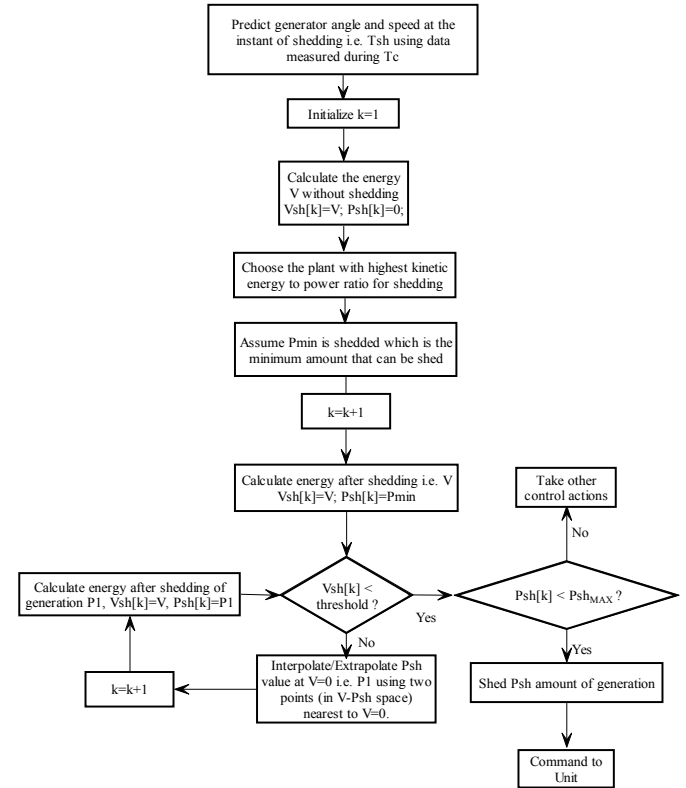


Fig. 1: Flowchart for generation shedding

After completion of the control action, the energy margin ($V(t) - V_{cr}$) is re-calculated. If it is unstable, further control action is required. In case very high amounts of generation are required to be shed to stabilize the system, there may be alternative emergency control strategies such as controlled islanding that may be more appropriate [20].

IV. RESULT

As the MOD of a particular fault location may change with change in operating condition, both MOD and $\overline{KE}$ are calculated for two different operating conditions (say at 10:00 AM and 10:30 AM). The method was tested on the IEEE 39 bus test system [21] shown in Fig. 2, wherein the variation in operating condition was obtained by adding a load of 400 MW at bus 22 and supported by generator 7. This is just to show how the method works, otherwise all the loads may vary with time in real system. The fault is considered to occur between 10:00-10:30 AM.

There are total fifteen MODs in the system for the two operating conditions. For example, for a fault on Line 1-2, the MOD is $M1 = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, and for a fault at bus 16, the MOD is $M2 = \{3, 4, 5, 6, 7, 8, 9\}$. For line faults, a single circuit line was replaced with a double circuit line of same equivalent impedance and the fault was cleared by removing the faulted circuit. For this system, all possible MODs at two operating conditions are shown in Table 2 along with the corresponding fault locations.

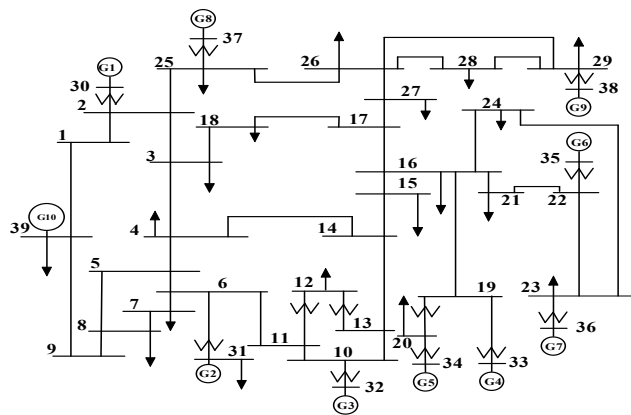


Fig.2: IEEE 39 bus test system

TABLE 2: MODS AND CORRESPONDING FAULT LOCATIONS

MOD	Fault Locations	
	Operating Condition 1	Operating Condition 2
M1={1 2 3 4 5 6 7 8 9}	Line: 1-2, 2-3	Line: 1-2, 2-3
M2={3 4 5 6 7 8 9}	Bus: 2, 16, 17, 18, 21, 24, 25, 30 Line: 3-18, 15-16, 17-18	Bus: 2, 16, 17, 18, 21, 24, 25, 30 Line: 3-18, 17-18, 17-27
M3={2 3 4 5 6 7 9}	Bus: 1, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 39 Line: 3-4, 4-5, 4-14, 5-6, 5-8, 6-7, 6-11, 7-8, 8-9, 10-13, 13-14, 14-15, 16-17, 17-27, 10-11	Bus: 1, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 39 Line: 3-4, 4-5, 4-14, 5-6, 5-8, 6-7, 6-11, 7-8, 8-9, 10-13, 13-14, 14-15, 10-11
M4={4 5 6 7}	Line: 16-19, 16-24	Line: 16-19
M5={4 5}	Bus: 19, 33	Bus: 19, 33
M6={6 7}	Bus: 22, 23, 35	Bus: 22, 23, 35
M7={1 8 9}	Line: 2-25	Line: 2-25
M8={0}	Line: 1-39, 9-39	Line: 1-39, 9-39
M9={3}	Bus: 32	Bus: 32
M10={5}	Bus: 20, 34	Bus: 20, 34
M11={7}	Bus: 36	Bus: 36 Line: 23-24, 22-23, 21-22, 16-24, 16-21, 16-17, 15-16
M12={8}	Bus: 37	Bus: 37
M13={9}	Bus: 26, 27, 28, 29, 38 Line: 25-26, 26-27, 26-28, 26-29, 28-29	Bus: 26, 27, 28, 29, 38 Line: 25-26, 26-27, 26-28, 26-29, 28-29
M14={4 5 6 7}	Line: 16-21, 21-22, 23-24	
M15={2}	Bus: 31	Bus: 31

For every fault location, MOD is identified and the $\overline{KE}$ calculated offline for two operating conditions and stored. It may be noted that for changes in fault location and post-fault topology, the MODs may or may not change, but the $\overline{KE}$ must always change. MODs and $\overline{KE}$ for two fault locations and operating conditions have been given in Table 3. Total number of stored fault cases was 146 i.e. 39 bus fault and 34 line fault for each of the two operating conditions.

TABLE 5: RANKING OF THE MODS IN ASCENDING ORDER OF EUCLIDEAN DISTANCE "D"

D	0.0103	0.0109	0.0149	0.0328	0.0358	0.0483	0.0566	0.0603	0.11	0.651	0.669	0.693	0.7	0.725
Fault location	Bus 29	Bus 29	Line 28-29	Line 28-29	Bus 28	Bus 38	Bus 28	Bus 38	Line 26-29	Line 17-27	Line 2-25	Line 2-25	Line 17-27	Bus 2
Operating condition	1	2	1	2	1	2	2	1	1	1	1	2	2	1
MOD	M13	M13	M13	M13	M13	M13	M13	M13	M13	M3	M7	M7	M2	M2

After M13, the next closer MODs are M3, M7, M2 etc.

TABLE 3: $\overline{KE}$ AND MOD FOR DIFFERENT FAULT LOCATIONS, CALCULATED OFFLINE USING SIMULATION DATA

Fault location	Operating Condition	$\overline{KE}$	MOD
Bus 29	1		M13={9}
	2		M13={9}
At 50% of Line 15-16	1		M2={3 4 5 6 7 8 9}
	2		M11={7}

With the offline look-up table built as described before, three test cases are simulated to represent scenarios where WAMS data for actual faults are used for transient stability assessment and control.

A. Fault on bus 29, MOD is M13 = {9}

Offline Computation:

MODs and $\overline{KE}$ for faults at different buses and lines are calculated offline and stored for two operating conditions along the daily load curve as explained before.

Online Computation:

If a fault occurs in the period between the two operating points, for which MODs were identified and stored, the correct MOD will be identified from the stored ones. Therefore, the method has to be tested for a new operating condition, that lies in between the two. Load at bus 22 was taken as 200 MW and this is considered to be the third operating condition.

step 1: Calculation of $\overline{KE}$: A balanced fault of duration 0.18s was created at bus 29. The $\overline{KE}$ gained by the generators in first 0.1s of the fault was calculated from the speeds measured by PMU which has been shown in Table 4.

TABLE 4: $\overline{KE}$ GAINED BY GENERATORS IN 0.1 S OF FAULT AT BUS 29, CALCULATED USING SPEEDS MEASURED WITH PMU

Fault location	$\overline{KE}$
Bus 29	

step 2: Ranking of Probable MODs- Using k-NN method, the MODs for fault cases similar to the actual fault, were detected and then ranked in ascending order of the Euclidean distance "D" from $\overline{KE}$ measured to $\overline{KE}$ stored for two operating conditions. First few ranked MODs are shown in Table 5.

step 3: Calculation of the Post Fault SEP- The post fault stable equilibrium point was calculated as per Step 1 of Sec. II-B.

$\theta^s = [-0.024 \ 0.204 \ 0.3033 \ 0.254 \ 0.38 \ 0.286 \ 0.55 \ 0.249 \ 0.46 \ -0.1591]$
 step 4: Calculation of the CUEP for Each MOD- For each MOD, corresponding CUEP was calculated as per Sec. II-A. As an example, the CUEP for Mod M13 is given below-

$\theta^u = [0.116 \ 0.317 \ 0.475 \ 0.549 \ 0.667 \ 0.593 \ 0.88 \ 0.52 \ 2.372 \ 0.0406]$
 step 5: Identification of the Correct MOD- For each MOD, the $(\frac{\Delta V_{pe}}{KE_{corr}})$ was calculated at the time of fault clearing. Minimum of $(\frac{\Delta V_{pe}}{KE_{corr}})$ indicates the correct MOD.

TABLE 6: IDENTIFICATION OF THE CORRECT MOD FROM THE RANKED MODS

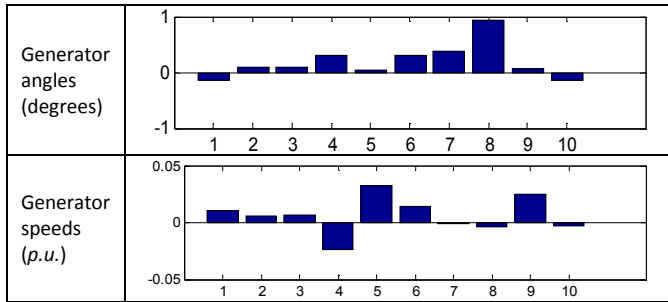
MOD	M13	M3	M7	M2
$\frac{\Delta V_{pe}}{KE_{corr}}$	0.8458	8.852	4.323	7.56

Table 6 shows that the correct MOD for a fault on bus 29 is M13 = {9}.

step 6: Assessment of first swing transient stability: Use the CUEP calculated in step 4 for MOD 13, to calculate the energy at fault clearing and the CUEP. In this case, V_{cl} is 4.979 and V_{cr} comes out to be 4.24. As $V_{cl} > V_{cr}$, the system will become unstable.

step 7: Prediction of Generator Angles and Speeds- As explained in Section III-B, generator angles and speeds have to be predicted at the instant of shedding using 0.05 s data after fault clearing. This instant is taken approximately as 0.2-0.3 s after fault clearing. The average of quadratic and cubic polynomial fit was selected as the prediction function. The error between the predicted and actual values of angle and speed after 0.2s of fault clearing is shown in Table 7 and found to be very less for all ten generators.

TABLE 7: ERROR BETWEEN PREDICTED AND ACTUAL VALUES



step 8: Estimation of Control Action Required- If energy margin is known, control actions can be taken like generation shedding (amount and location), dynamic braking (resistance value, location and duration of on time), fast valving (location, duration and percentage of opening) etc. In this work, generation shedding has been used as transient stability control.

Generation was shed from plant 9 after 0.2 s of fault clearing. Amount of generation shedding required to stabilize the system was calculated from TEF which was further verified using repeated time domain simulations (TDS). Results are given in Table 8.

TABLE 8: AMOUNT OF GENERATION SHEDDING REQUIRED

Fault Duration (s) (CCT=0.173)	Amount of Shedding Required (in percentage of 9 th generator output)	
	TEF	TDS
0.174	11%	1%
0.180	32%	22%

0.184	49%	40%
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It can be seen that TEF provides result with sufficient accuracy. Though CCT for fault at bus 29 is 0.173s, TEF gives CCT as 0.164s that causes an inherent error in calculating control actions required. If the fault duration is in between 0.164 s and 0.173 s, TEF will show the system as unstable though the system is actually stable. As TEF gave pessimistic results in CCT, results in calculation of control actions required like generation shedding was also pessimistic. It may be noted that generation shedding is done by tripping single or multiple units. Therefore error of few percentage is acceptable. In practice, dynamic braking and other lighter control actions may be used when TEF shows the system as marginally unstable.

B. Fault on bus 15: MOD is M3 = {2, 3, 4, 5, 6, 7, 9}

This is an example of inter area mode separation after a fault. Generator 10 has very high inertia as it is the aggregated model of a group of generators. Generator 1 and 8 are strongly coupled with the 10th generator. Therefore, after a fault at bus 15, rest of the generators tends to be out of step. Fault duration was taken as 0.287s while the CCT computed using time domain simulation was 0.267s.

I. Offline Computation:

It is similar to the previous case.

II. Online Computation:

Loads are assumed to be the same as given in Sec. IV-A.

Steps 1-4 can be done similar to the Sec. IV-A.

step 5: Identification of Correct MOD- For each MOD, $(\frac{\Delta V_{pe}}{KE_{corr}})$ is calculated after fault clearing. Minimum of $(\frac{\Delta V_{pe}}{KE_{corr}})$ indicates the correct MOD. Table 9 shows that the correct MOD is M3 = {2, 3, 4, 5, 6, 7, 9}.

TABLE 9: IDENTIFICATION OF THE CORRECT MOD FROM THE RANKED MODS

MOD	M3	M2	M11	M5	M13
$\frac{\Delta V_{pe}}{KE_{corr}}$	0.995	1.05	1.52	2.6	3.16

step 6: Assessment of first swing transient stability: Use the CUEP calculated in step 4, to calculate the energy at the instant of fault clearing and the CUEP. V_{cl} is 12.47 and V_{cr} comes out to be 11.65. As $V_{cl} > V_{cr}$, the system will become unstable.

step 7: Prediction of Generator Angles and Speeds- Generator angles and speeds were predicted at the instant of generation shedding i.e. 0.2 s after fault clearing.

step 8: Estimation of Control Action Required- Generation was shed from plant 7 as it was having highest kinetic energy/MW ratio and hence less amount generation would be shed. Amount of generation shedding required as obtained using the proposed TEF technique was 55%. It was verified using repeated time domain simulations (TDS) with different amount of generation shedding that comes out to be 50% which is very close to the result obtained from TEF.

C. Fault at 50% of line 1-39: MOD={0}

This is a very rare case of no separation even after sustained fault. Kinetic energy gained by the generators during fault is such that all the generators accelerate together.

Offline Computation:

It is similar to the previous case presented in Sec. IV-A.

Online Computation:

Loads are assumed to be the same as given in Sec. IV-A. Steps 1-3 can be done similar to the Sec. IV-A.

step 4: Calculation of the CUEP for Each MOD- CUEP is calculated for all MOD except MOD={0}.

step 5: For first few MODs ranked in ascending order of the distance "D", The $(\frac{\Delta V_{pe}}{KE_{corr}})$ was calculated at the time of fault clearing. If no generator is going to be out of step, KE_{corr} would be zero. Therefore, the $(\frac{\Delta V_{pe}}{KE_{corr}})$ was not calculated for MOD = {0}. The minimum of $(\frac{\Delta V_{pe}}{KE_{corr}})$ in Table 10 indicates the correct MOD which is $M_1 = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$.

TABLE 10: IDENTIFICATION OF THE CORRECT MOD

MOD	M8	M1	M3	M7	M2
dV_{pe}/KE_{corr}	-	47.8076	65.896	169.748	59.846

step 6: Use the CUEP calculated in step 4 for MOD M1, to calculate the energy at the fault clearing instant and the CUEP. $\theta^u = [1.07 \ 1.677 \ 1.824 \ 1.716 \ 1.831 \ 1.772 \ 2.083 \ 1.479 \ 1.88 \ -0.95]$ V_{cl} is 0.2695 and V_{cr} comes out to be 14.995. As $V_{cl} < V_{cr}$, the system will remain stable and no control action is required.

D. Other Faults:

The method was tested for faults at different buses and on different lines. Results are given in Table 11. For line faults, the fault was created at 25% of the line selected from the double circuit line and that line was tripped to clear the fault. For bus fault, no line was tripped.

TABLE 11: ESTIMATION OF THE GENERATION SHEDDING REQUIRED FOR FAULTS AT DIFFERENT LOCATIONS

Fault Location	CCT (s) TEF,TDS	Fault duration	MOD	Generation shedding required in percentage of the output of shedded generator		
				Gen No	TEF	TDS
Line 2-3	0.421, 0.417	0.43	M1	7	23%	29%
Bus 2	0.317, 0.319	0.33	M2	7	26%	28%
Line 16-19	0.27, 0.276	0.29	M4	7	27%	20%
L10-11	0.319, 0.312	0.32	M3	7	32%	37%
L6-11	0.33, 0.332	0.34	M3	7	40%	32%
36	0.165, 0.18	0.19	M11	7	100%	85%
34	0.26, 0.265	0.28	M10	5	100%	100%
37	0.277, 0.283	0.29	M12	8	100%	100%

E. Demonstration using Time Domain Simulation:

The method has been demonstrated for the first fault case (described in Sec. IV-A) using time domain simulation in Fig. 3. It can be seen that without control action the system becomes unstable after fault clearing whereas the system remains stable after 49% generation shedding at 9th generator. In practice, generation shedding is to be done not by certain percentage, but single or multiple units is to be tripped.

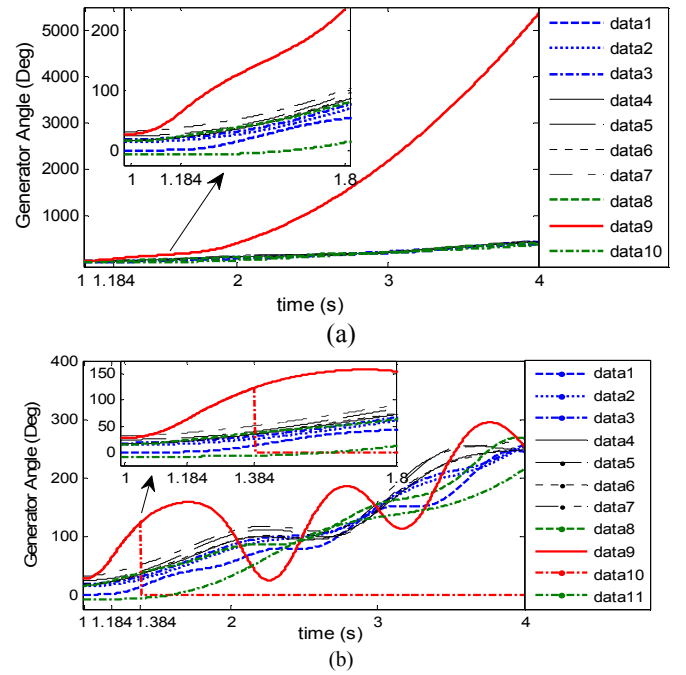


Fig. 3: Fault occurs on bus 29 at 1 s, cleared at 1.184 s. a) no control action, system instability b) 49% generation shed at 9th Gen at 1.384 s, stable case.

It may be noted that TEF is not as accurate as time domain simulation. If the fault duration is in between 0.164 s (CCT obtained from TEF) and 0.173 s (CCT obtained from TDS), TEF will show the system as unstable though the system is actually stable. But the energy margin (represents fault severity) obtained from TEF will show a smaller value and hence a small amount of generation shedding required. In that case, smaller control actions like dynamic breaking may be applied instead of tripping that unit. The biggest advantage of using TEF is that it can give indication of fault severity, depending on which amount and type of control action can be decided.

V. ADVANTAGES

Advantages of the method over the recently published machine learning techniques based on time domain simulations [1]-[4] have been provided below:

1. Machine learning based techniques use time domain simulations of at least 3 seconds. In TEF, BCU methods need simulation of faulted network of smaller duration, typically 0.4-0.5 s, to generate offline dataset.

2. In machine learning techniques, for a particular fault location, simulations are to be done for different fault durations, resistances and types. In TEF, for a particular location, MOD is to be calculated only once.

The above two advantages can significantly reduce the offline database generation time.

3. If the system topology changes, the data set generated for machine learning techniques, has to be regenerated. Whereas for TEF, fault cases most affected by topology changes, can be known from sensitivity techniques [22]. Therefore, only few fault cases are to be regenerated.

4. Machine learning based techniques normally use time domain simulation which may not be able to provide qualita-

1 tive results. Hence the amount of control actions required cannot
2 be obtained from the machine learning based techniques.

3 5. The CCT for a particular fault location changes widely
4 with change in loading condition. But, the MOD for a particu-
5 lar fault location remains unchanged for small load variations.
6 Hence data set is to be generated less frequently.

7 VI. CONCLUSION

8 This paper proposes a novel technique to apply Transient
9 Energy Function method for assessment of transient stability
10 in real time. One of the reasons for not applying TEF method
11 in real time in past, is difficulties associated with calculation
12 of CUEP in real time [15]. The proposed method uses a com-
13 bination of BCU method (offline) and the MOD technique
14 (online) to avoid computation of CUEP using BCU method in
15 real time. Using the BCU method, the MOD is identified for
16 fault at each location of the grid and stored offline. After an
17 actual fault at any location, the $\overline{KE}$ gained by the generators is
18 calculated from PMU data and the most probable MODs are
19 ranked on the basis of Euclidean distance between the meas-
20 ured $\overline{KE}$ and the stored $\overline{KE}$ s. Then, using $(\frac{\Delta V_{pe}}{KE_{corr}})$, the correct
21 MOD is identified. With knowledge of correct MOD, tripped
22 line after fault, pre-fault topology and loading condition, the
23 transient stability status and energy margin has been predicted
24 to estimate the control action required in unstable cases. As it
25 is difficult to use time domain simulation in real time and
26 does not give severity of the fault and the machine learning
27 based techniques have many limitations, this method provides
28 an alternative to assess and control transient stability in real
29 time. The main contribution of this work is calculating the
30 CUEP in real time which will allow to use TEF in real time.

31 VII. REFERENCE

32 [1] N. Amjady and S. F. Majedi, "Transient stability prediction by a hybrid
33 intelligent system," *IEEE Trans. Power Syst.*, vol. 22, no. 3, pp. 1275-
34 1283, August 2007.
35 [2] T. Amraee and S. Ranjbar, "Transient Instability Prediction Using Deci-
36 sion Tree Technique", *IEEE Trans. Power Syst.*, Vol. 25, No. 3, pp.
37 3028-3037, 2013.
38 [3] F. R. Gomez, A. D. Rajapakse, U. D. Annakkage and I. T. Fernando
39 "Support vector machine based algorithm for post-fault transient stabil-
40 ity status prediction using synchronized measurements", *IEEE Trans.
41 Power Syst.*, vol. 26, no. 3, pp.1474-1483, 2011.
42 [4] K. Mei and S. M. Rovnyak "Response-based decision trees to trigger
43 one-shot stabilizing control", *IEEE Trans. Power Syst.*, vol. 19, no. 1,
44 pp. 531-537, 2004.
45 [5] T. Athay, P. Podmore, and S. Virmani, "A practical method for the
46 direct analysis of transient stability," *IEEE Trans. Power Apparatus and
47 Systems*, vol. 98, no. 2, pp. 573-584, Mar. 1979.
48 [6] A. A. Fouad, K. C. Krumpel, K. R. C. Mamandur, S. E. Stanton, M. A.
49 Pai, and V. Vittal, "Transient Stability Margin as a Tool for Dynamic
50 Security Assessment," *EPR Report EL-1755*, March, 1981.
51 [7] A. A. Fouad, and S. E. Stanton, "Transient Stability of a Multimachine
52 Power System. Part I: Investigation of the System Trajectory. Part II:
53 Critical Transient Energy," *IEEE Trans. Power Apparatus and Systems*,
54 vol. 100, pp. 4308-4324, 1981.
55 [8] H. D. Chiang, F. Wu and P. Varaiya, "A BCU method for direct analysis
56 of power system transient stability," *IEEE Trans. Power Syst.*, vol. 9, pp.
57 1194-1208, Aug. 1994.
58 [9] N. Kakimoto, Y. Ohsawa and M. Hayashi, "Transient Stability Analysis
59 of Multi-Machine Power Systems with Field Flux Decays Via Lyapunov's
60 Direct Method," *IEEE Trans. Power Apparatus and Systems*, Vol.
99, No. 5, pp. 1819-1827, 1980.

[10] T. Athay and D. I. Sun, "An Improved Energy Function for Transient
Stability Analysis," *Proc. of International Symposium on Circuits and
Systems*, Chicago, April 1981.
[11] A. R. Bergen and D. J. Hill, "A Structure Preserving Model for Power
System Stability Analysis," *IEEE Trans. Power Apparatus and Systems*,
Vol. 100, pp. 25-35, Jan. 1981.
[12] M. A. Pai, K. R. Padiyar and C. Radhukrishna, "Transient Stability
Analysis of Multi-Machine AC/DC Power Systems Via Energy-
Function Method," *IEEE PES Summer Meet.*, Portland, Oregon, 1981.
[13] A. A. Fouad and V. Vittal, "Power System Transient Stability Analysis
Using the Transient Energy Function Method," Prentice Hall, New Jer-
sey, 1992.
[14] P. W. Sauer and M. A. Pai, "Power System Dynamics and Stability,"
Prentice Hall, New Jersey, 1998.
[15] A. Llamas, J. De La Ree Lopez, L. Mili, A. G. Phadke and J. S. Thorp,
"Clarifications on the BCU method for transient stability analy-
sis," *IEEE Trans. Power Systems*, vol. 10, pp.210-219, 1995.
[16] N. Kakimoto, Y. Ohsawa, and M. Hayashi, "Transient Stability Analysis
of Electric Power System Via Lure' Type Lyapunov Functions, Part I
and II," *Trans. IEEE of Japan*, Vol. 98, No. 616, May/June, 1978.
[17] P. Bhui and N. Senroy, "Online Identification of Tripped Line for Tran-
sient Stability Assessment," accepted *IEEE Trans. Power Syst.*, May
2015.
[18] P. Bhui and N. Senroy, "Application of Recurrence Quantification
Analysis to Power System Dynamic Studies," accepted *IEEE Trans.
Power Syst.*, Feb 2015.
[19] M. Takahashi, K. Matsuzawa, M. Sato, K. Omata, R. Tsukui, T. Naka-
mura, and S. Mizuguchi "Fast generation shedding equipment based on
the observation of swings of generators," *IEEE Trans. Power Syst.*,
vol.3, no.2, pp.439-446, 1988.
[20] N. Senroy, G. T. Heydt and V. Vittal, "Decision tree assisted controlled
islanding," *IEEE Trans. Power Syst.*, vol. 21, no. 4, pp. 1790-1797,
2006.
[21] M. A. Pai, *Energy Function Analysis for Power System Stability*, 1989,
Kluwer.
[22] V. Chadalavada and V. Vittal, "Transient stability assessment for net-
work topology changes: Application of energy margin analytical sensi-
tivity", *IEEE Trans. Power Syst.*, vol. 9, no. 3, pp.1658-1664, 1994.
[23] A. A. Fouad, A. Ghafurian, K. Nodehi and Y. Mansour, "Calculation of
generation-shedding requirements of the B. C. Hydro system using tran-
sient energy function method", *IEEE Trans. Power Syst.*, vol. 1, no. 2,
pp.17-24, 1986.
[24] G. G. Karady and J. Gu, "A hybrid method for generator tripping," *IEEE
Trans. Power Syst.*, vol. 17, no. 4, pp.1102-1107, 2002.
[25] G. Trudel, S. Bernard and G. Scott, "Hydro-Québec's defence plan
against extreme contingencies," *IEEE Trans. Power Syst.*, vol. 8, no.
2, pp.445-451, 1993.
[26] H-C. Chang and H-C. Chen, "Fast Determination of Generation-
Shedding in Transient Emergency State", *IEEE Trans Energy Conv.*,
vol. PWRS8, pp.178-183, June 1993.
[27] M. Djukanovic, D.J. Sobajic and Y.H. Pao, "Neural-Net Based Deter-
mination of Generator Shedding Requirements in Electric Power Sys-
tems", *IEE Proc.-C*, vol. 139, pp.427-436 1992.
[28] Friedman, J. H., J. Bentley, and A. R. Finkel, "An Algorithm for Finding
Best Matches in Logarithmic Expected Time," *ACM Trans. Mathemat-
ical Software* 3, 209, 1977.
[29] J. Geeganage, U. D. Annakkage, T. Weekes and B. A. Archer, "Applica-
tion of Energy-Based Power System Features for Dynamic Security As-
sessment," *IEEE Trans. Power Syst.*, Vol. 30, No. 4, pp. 1957-1966, July
2015.
[30] N. Yorino et. al. "An Application of Critical Trajectory Method to BCU
Problem for Transient Stability Studies," *IEEE Trans. Power Syst.*, Vol.
28, No. 4, pp. 4237-4244, 2013.

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